



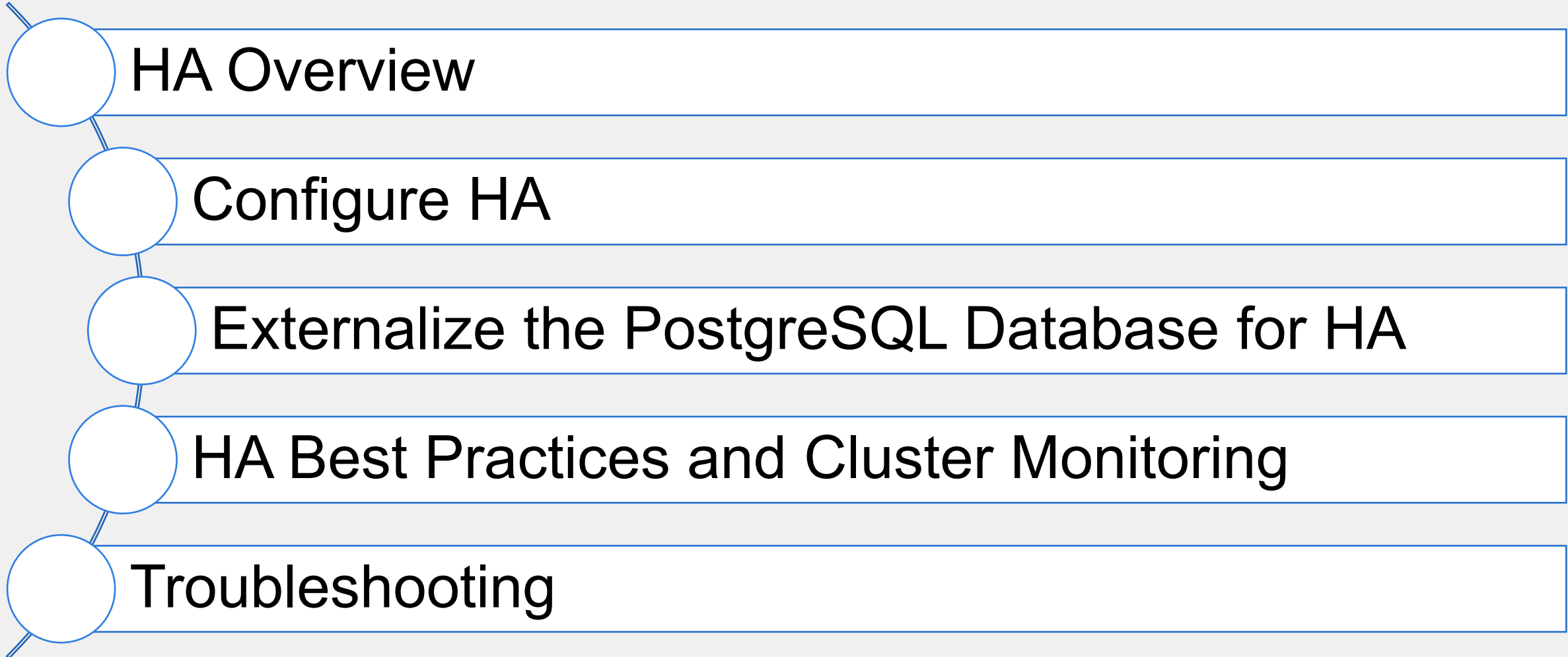
FORTINET®
Training Institute



FortiSOAR Administrator

High Availability

Lesson Overview



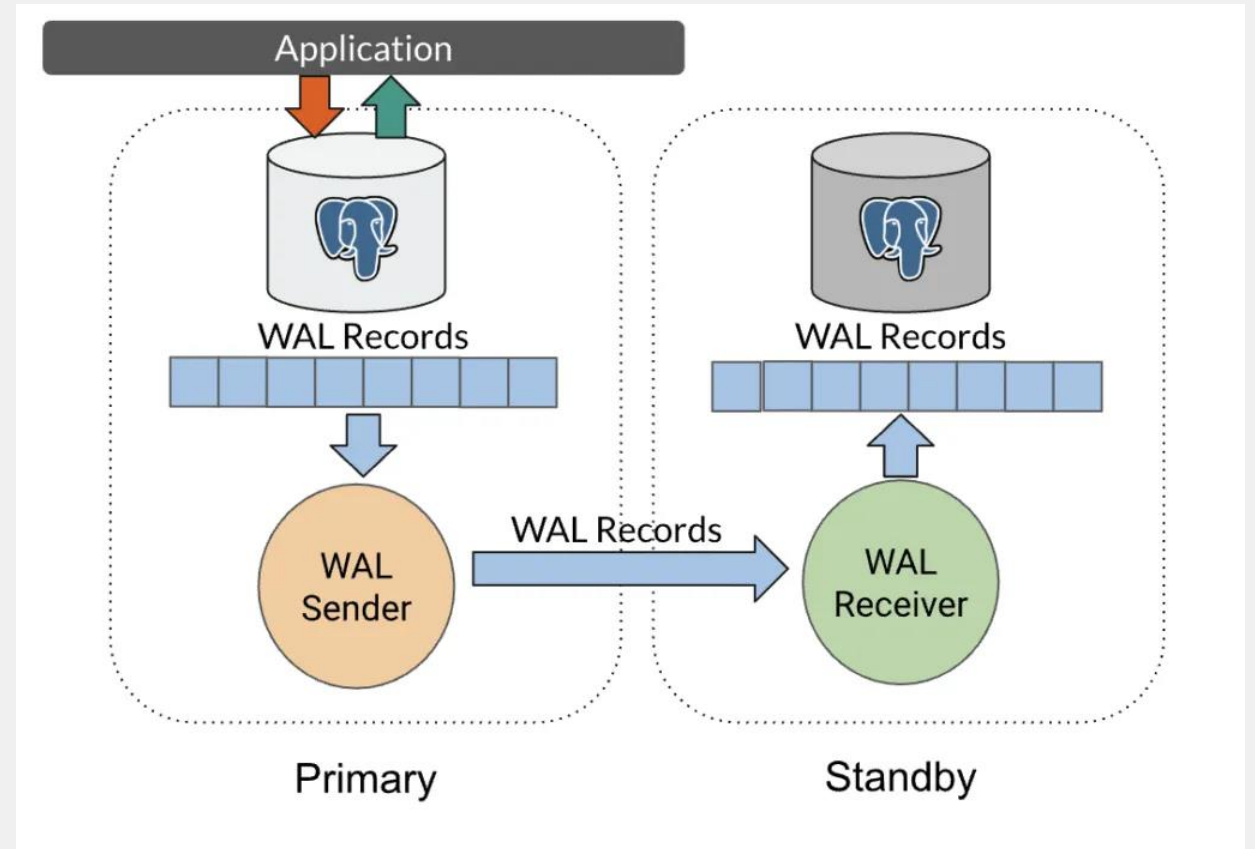
HA Overview

Objectives

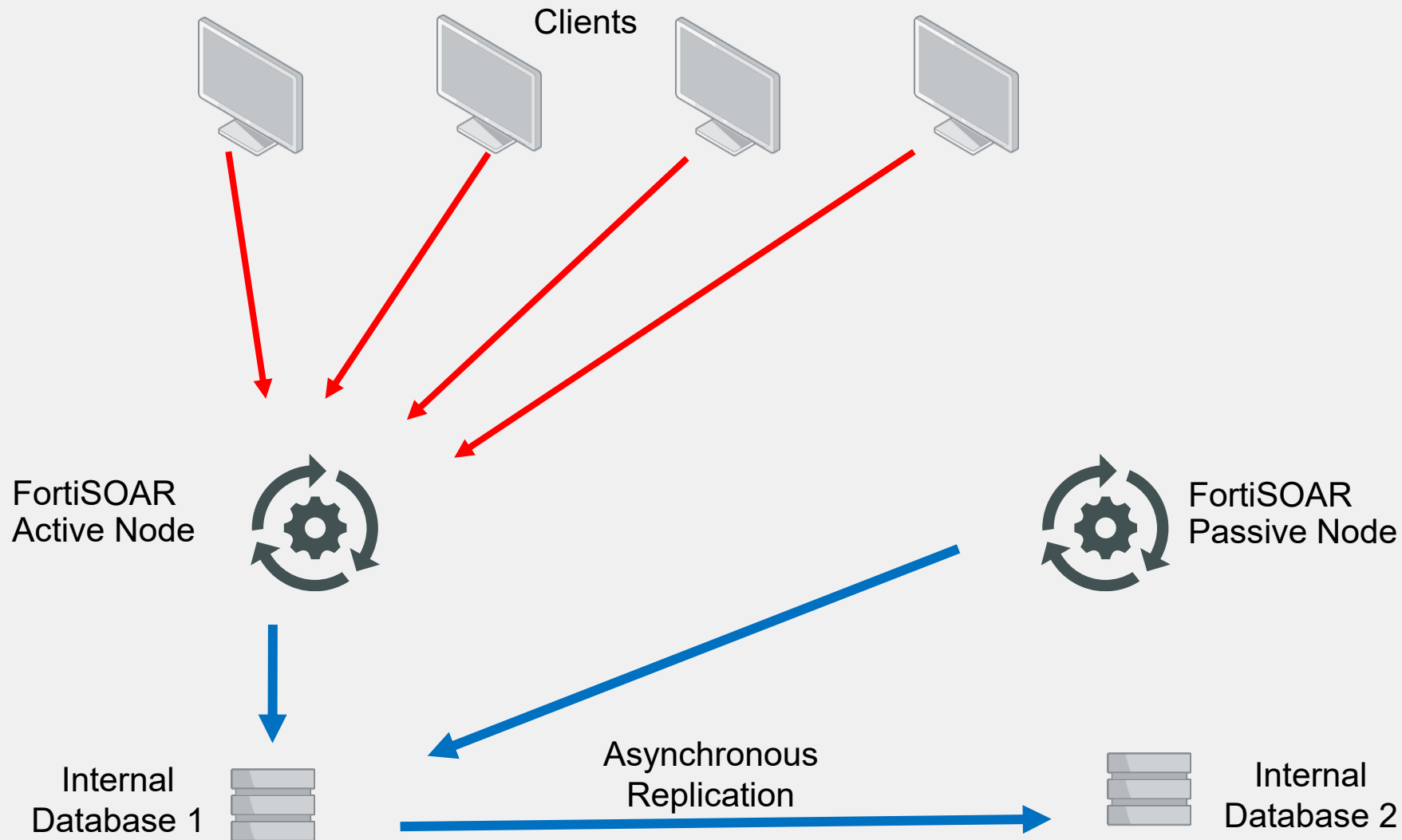
- Describe different HA implementation options
- Manage cluster licensing

HA Overview

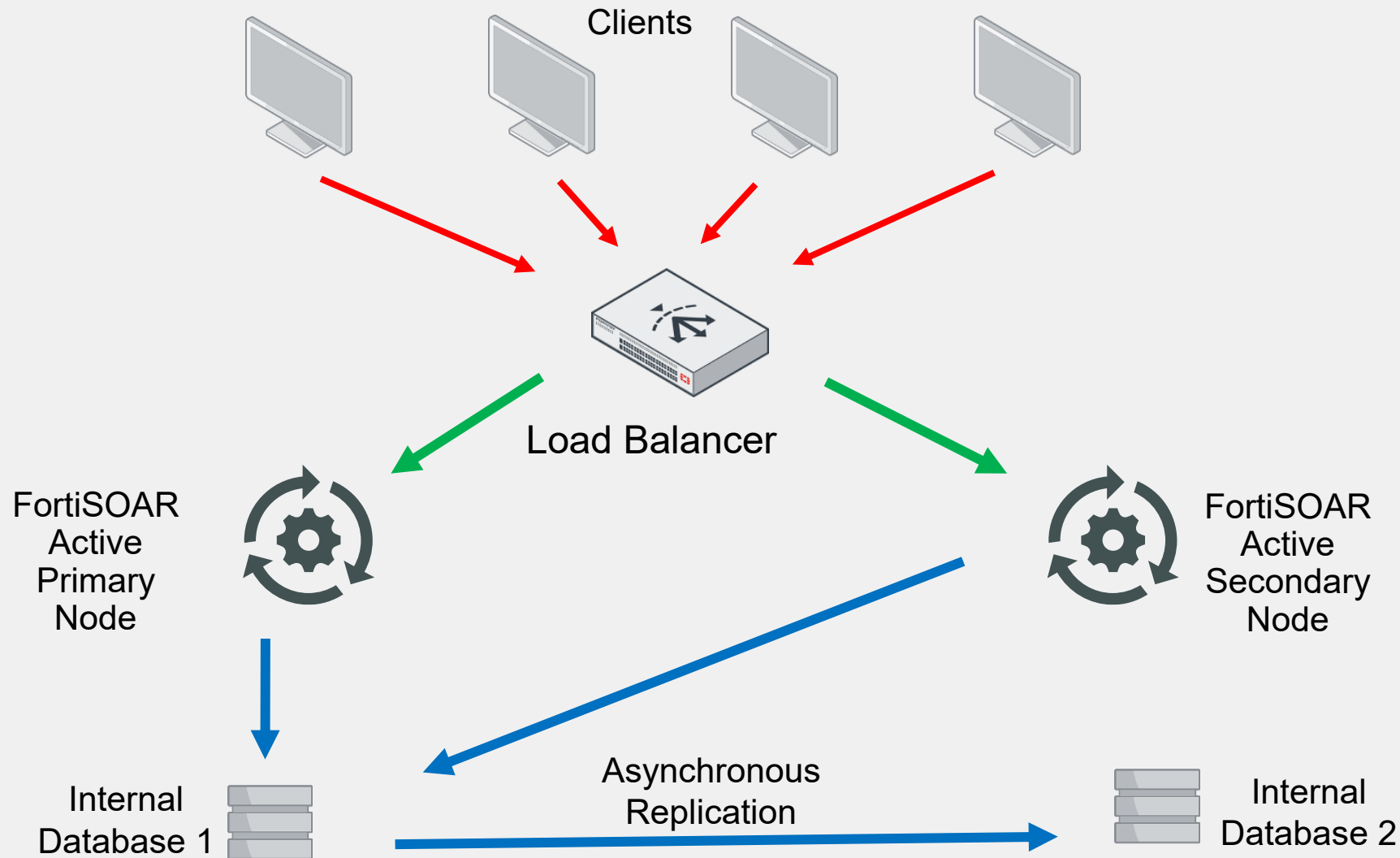
- Fastest recovery time:
 - FortiSOAR HA native clustering
- Other HA options with fast recovery time objectives:
 - Nightly database backups and incremental VM snapshots
 - HA provided by the underlying virtualization platform
 - External database
- FortiSOAR implements HA clustering with the use of PostgreSQL database clustering
 - Supports active-active (A-A) and active-passive (A-P) configurations
 - With both internal and external PostgreSQL databases
- You can use the FortiSOAR HA cluster for the following purposes:
 - Disaster recovery (DR)
 - Scaling



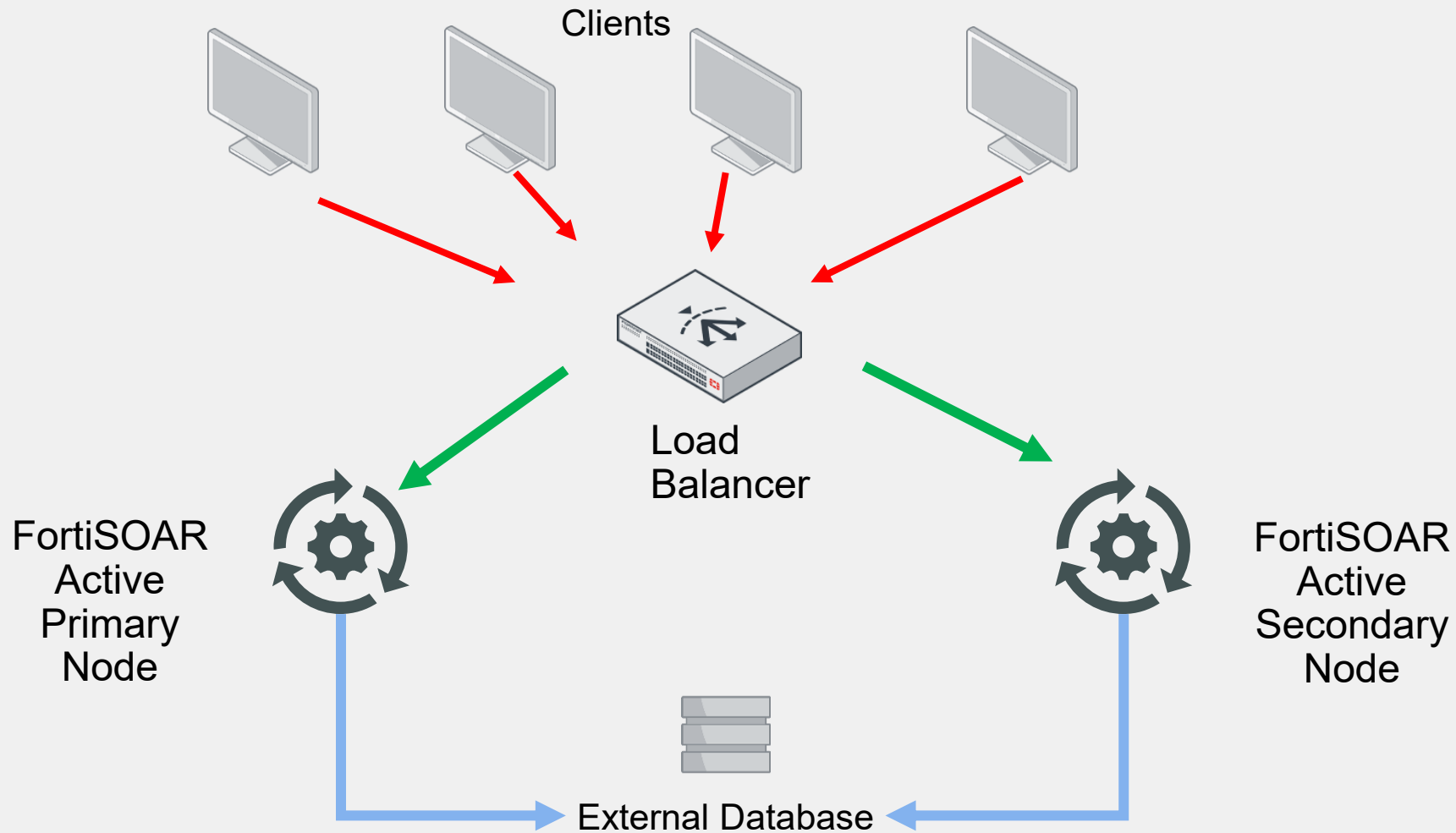
Active-Passive Cluster With Internal Database



Active-Active Cluster With Internal Database

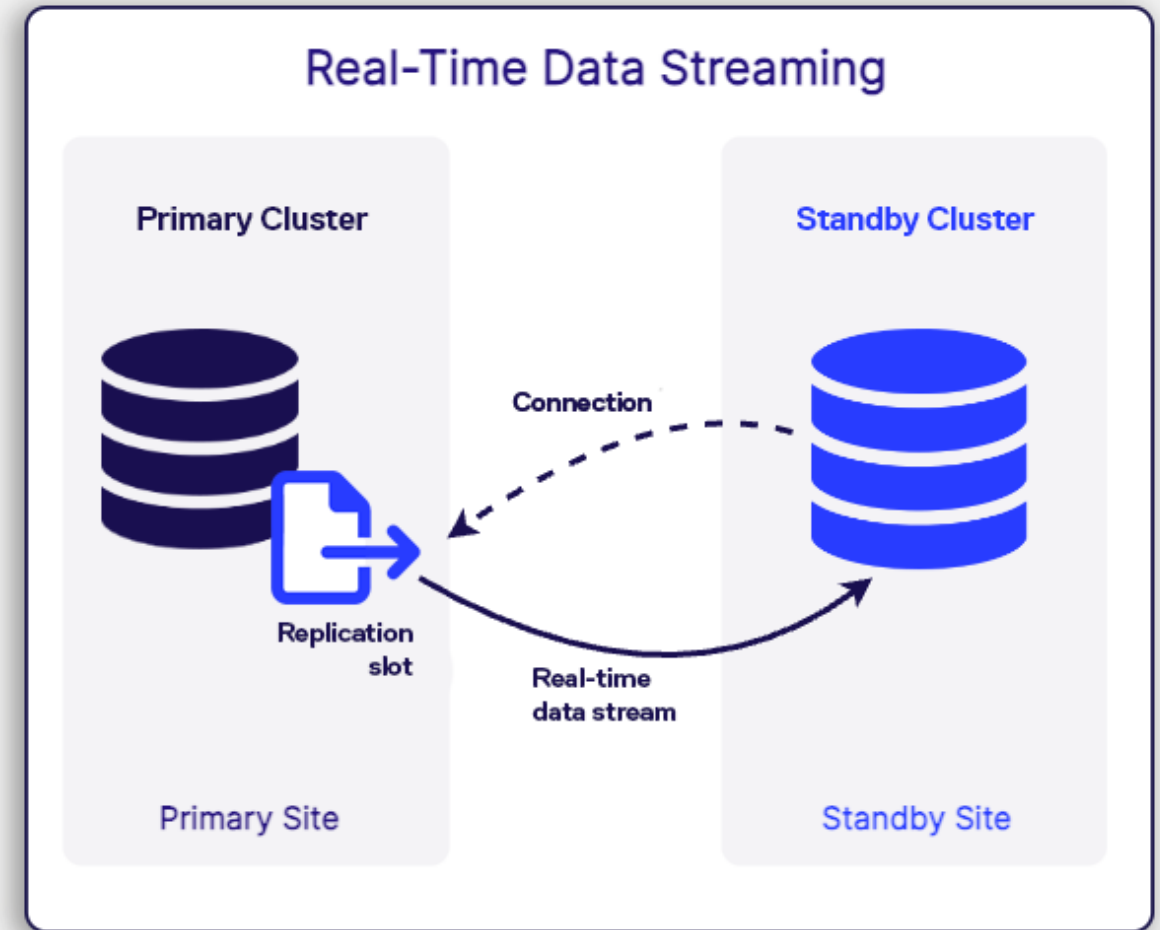


HA With an External PostgreSQL Database



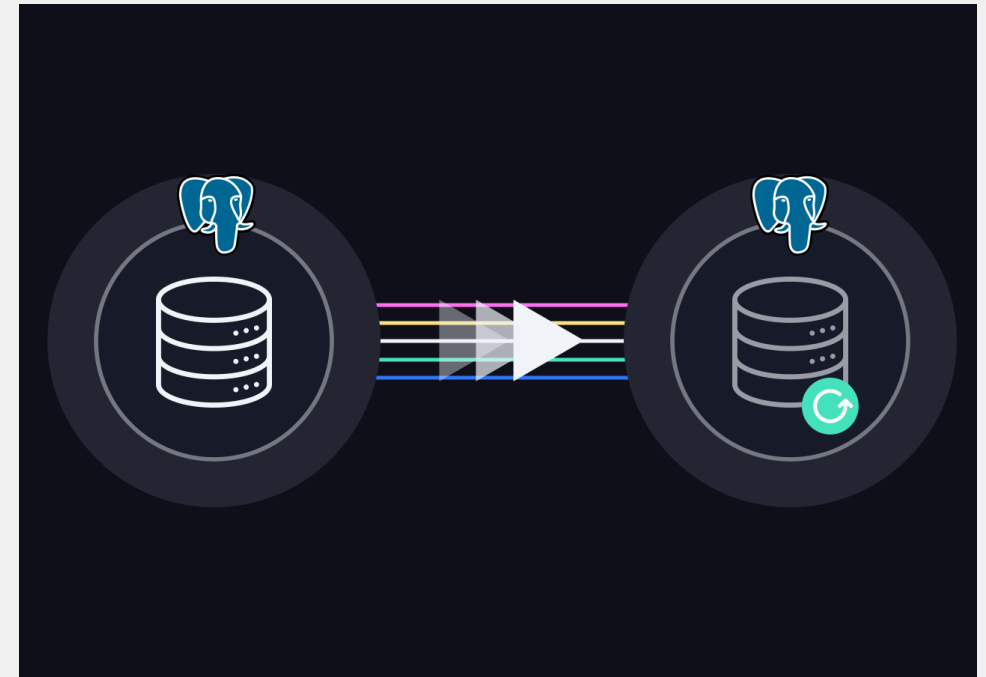
Primary Node Overview

- An HA cluster can only have one active primary node
- Secondary nodes can be active or passive
- For an internal database, all active nodes talk to the primary for all read/write operations
 - They are in read-only mode and replicate to the primary
- Starting from 7.2.0, replication slots add support for differential synchronization between primary node and secondary nodes
- These functions run only on the primary node:
 - Workflow Scheduler
 - Data indexing for quicksearch into Elasticsearch
 - Listeners configured for notifications
- If the active primary node goes down, another node must be promoted



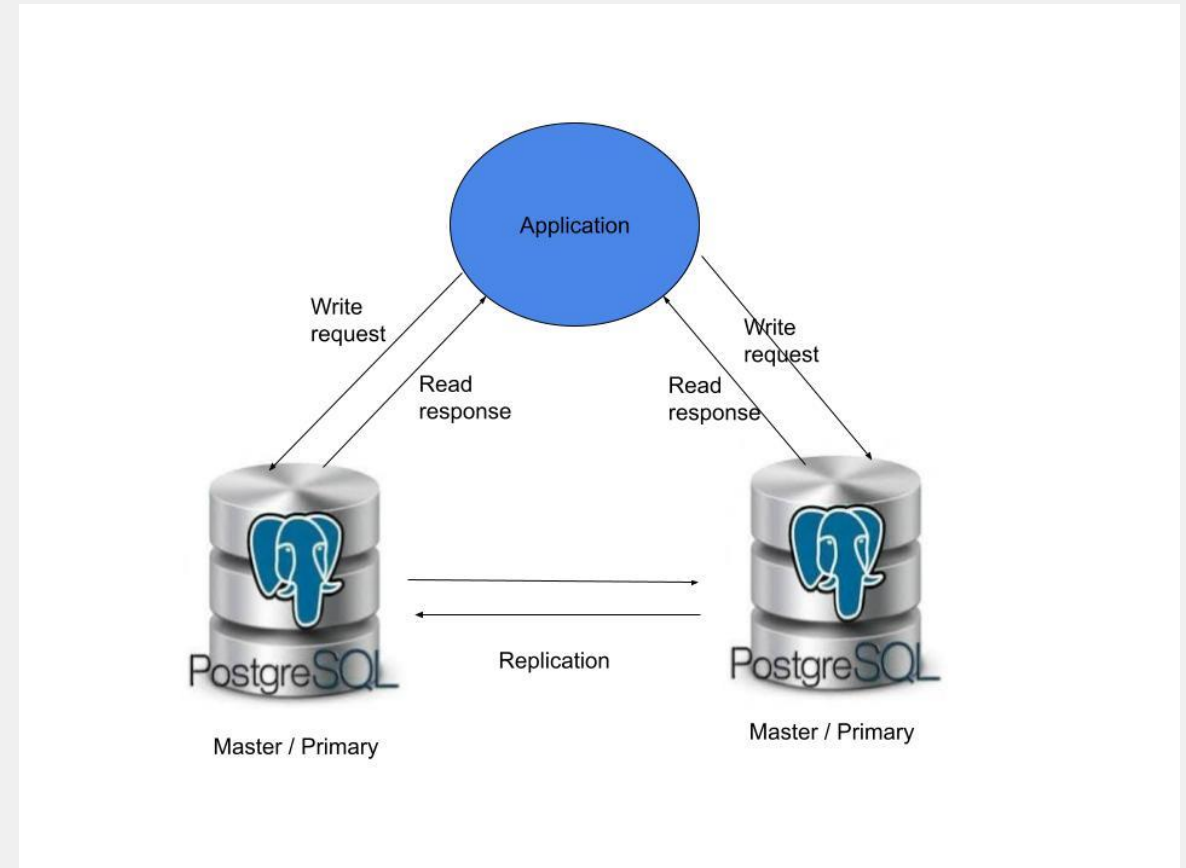
Custom Connectors in an HA Cluster

- Prior to 7.2.0, you had to upload the `TGZ` file of the custom connector *on all the nodes within the HA cluster* and manually install it using the CLI
 - Must have the same version of the connector on all nodes
 - The **Delete all existing versions** option must be selected when uploading
- Starting from 7.2.0, the installation process automatically installs the following connectors on other nodes, as if they were FortiSOAR-published connectors:
 - Custom connectors
 - Older versions of connectors that did not have their RPM available on the FortiSOAR server
 - Connectors that were created and published using the **Create New Connector** wizard
- Starting from 7.2.1, once you have installed a connector dependency on a FortiSOAR node using the **Install** link on the connector's configurations dialog, then that dependency is installed on other nodes



Internal PostgreSQL database

- FortiSOAR HA is based on internal clustering that takes care of replicating data (PostgreSQL) to all cluster nodes
- Administration CLI access is available for managing the cluster and performing the takeover (failover) operation, if necessary
- A-A HA cluster
 - At least two nodes are actively running the same kind of service at the same time
 - Achieve load balancing and horizontal scaling
 - Recommended to use proxy or load-balancer to effectively direct requests to all nodes
- A-P HA cluster
 - One or more passive or standby nodes are available to take over if the primary node fails
 - One active node and one or more passive nodes configured in a cluster, which provides redundancy



Cluster Licensing

- Additional user entitlement does not need to be the same across all nodes
- User count entitlement is always validated from the primary node
- During a takeover operation, the new primary node will swap licenses with the old primary node
- The **License Manager** page also displays the information about the other nodes in the cluster

Settings > License Manager

License Manager

Type

Perpetual

Edition

Enterprise

Total Users

2 Users

Nodes

Node Name	Status	Role	License Details
primary.internal.lab	Active	Primary	Serial Number: FSRVMPTM20000484 Total Users: 2 Device UUID: 520e5f3badcd9647b829c8905fec465b
secondary.internal.lab	Passive	Secondary	Serial Number: FSRVMPTM21000105 Total Users: 2 Device UUID: 927ee24c7eabd15e684c996bd04ea968

Update License

Update License

Knowledge Check

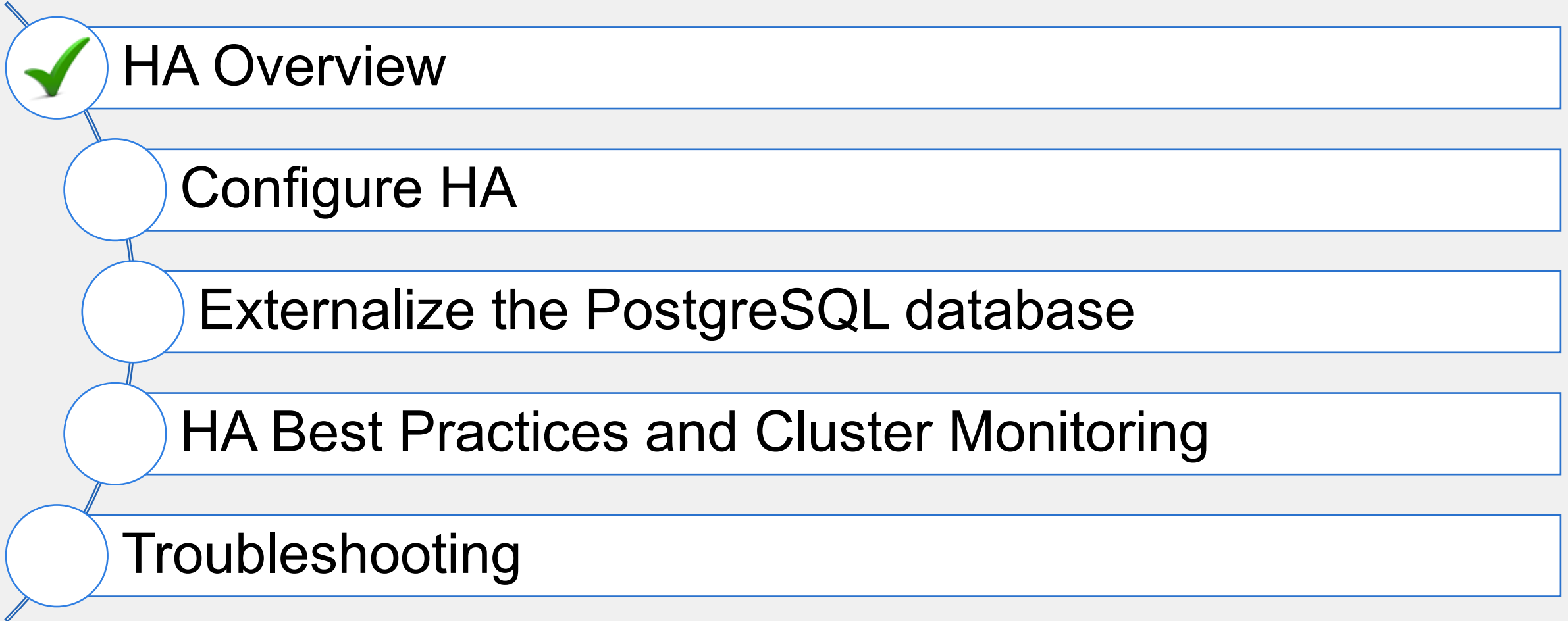
1. For scaling workflow execution across multiple nodes, which type of FortiSOAR HA clustering set up is recommended?

- ✓ A. Active-Active
- B. Active-Passive

2. Which statement about a FortiSOAR primary node is true?

- A. Multiple primary nodes can be set up to provide load balancing.
- ✓ B. There can be only one primary node in a cluster.
- C. The primary node can be passive or active.

Lesson Progress



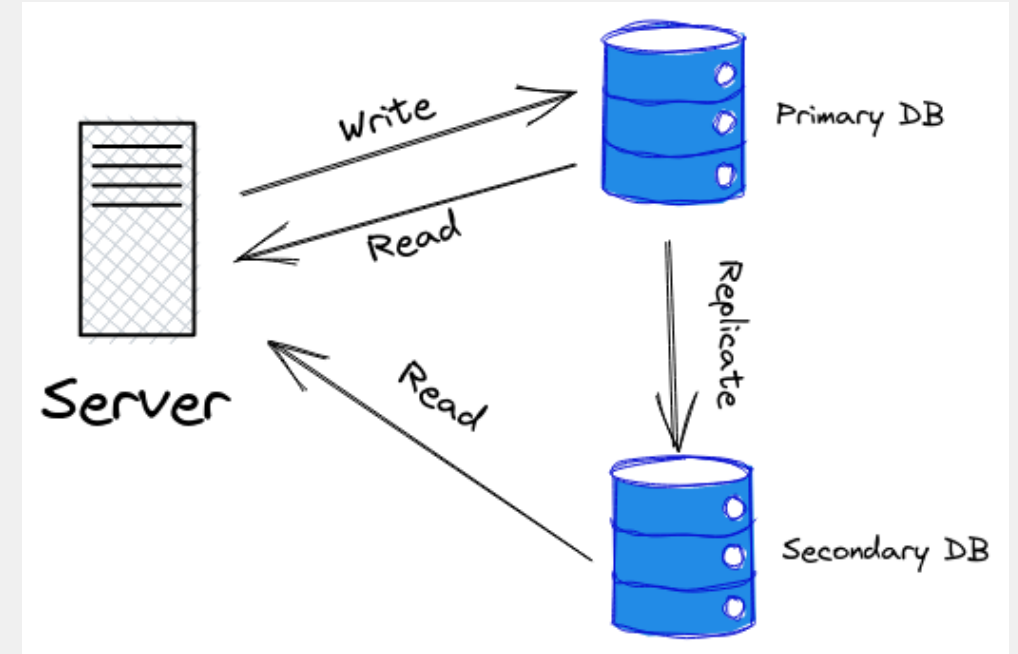
Configure HA

Objectives

- Identify the prerequisites for configuring HA
- Configure HA with an internal PostgreSQL database
- Configure HA with an external PostgreSQL database
- Understand the uses of the `csadm ha` command

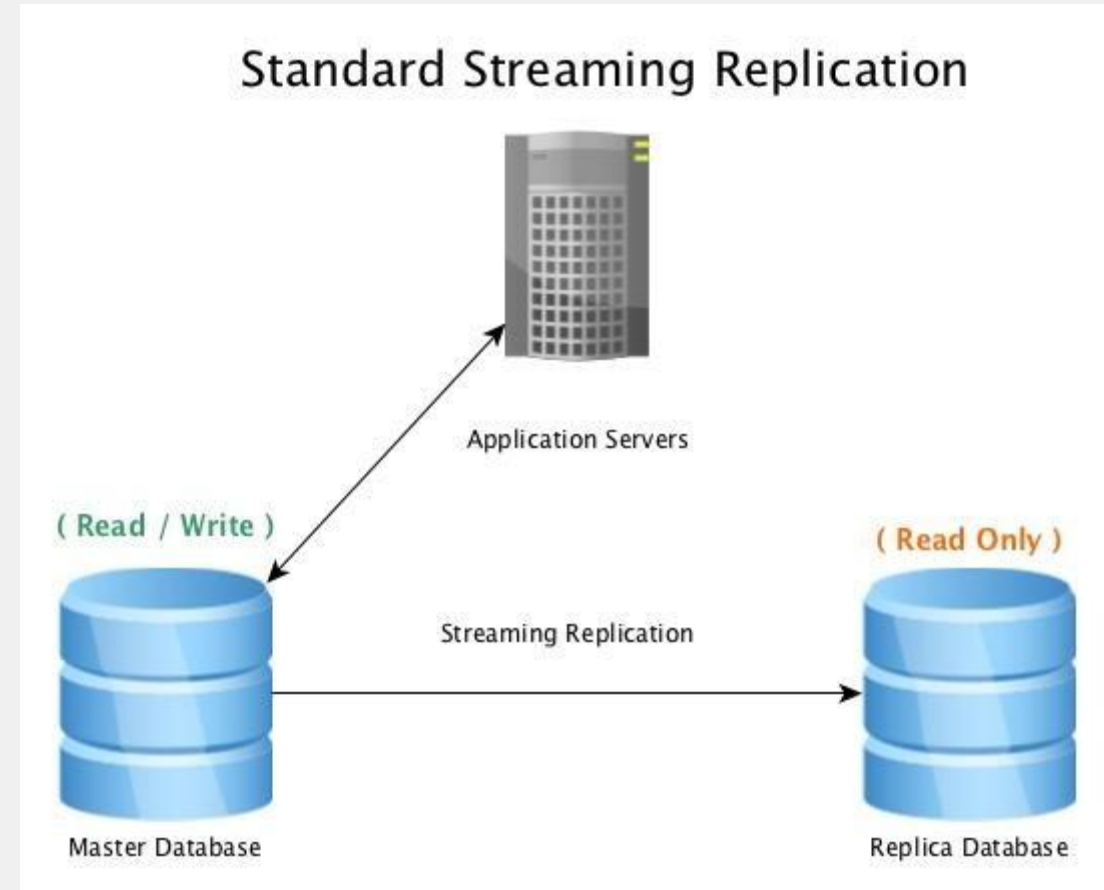
Prerequisites to Configuring HA

- All nodes are on the same version
- All nodes are resolvable through DNS
- SSH session must not time out:
 - Increase session timeout value in your SSH application, or
 - Use the `screen` (prior to 7.3.0) or `tmux` (7.3.0 and above) command to maintain the session
 - See the *Handle session timeouts while running the FortiSOAR upgrade* article in the Fortinet Knowledge Base
- The following ports must be accessible between HA nodes:
 - 5432 (PostgreSQL), 6379 (Redis), 9200 (ElasticSearch)
- Recommended: Installing the cluster behind a load balancer to ensure cluster address remains unchanged after failover
- Configuring HA is done through CLI only



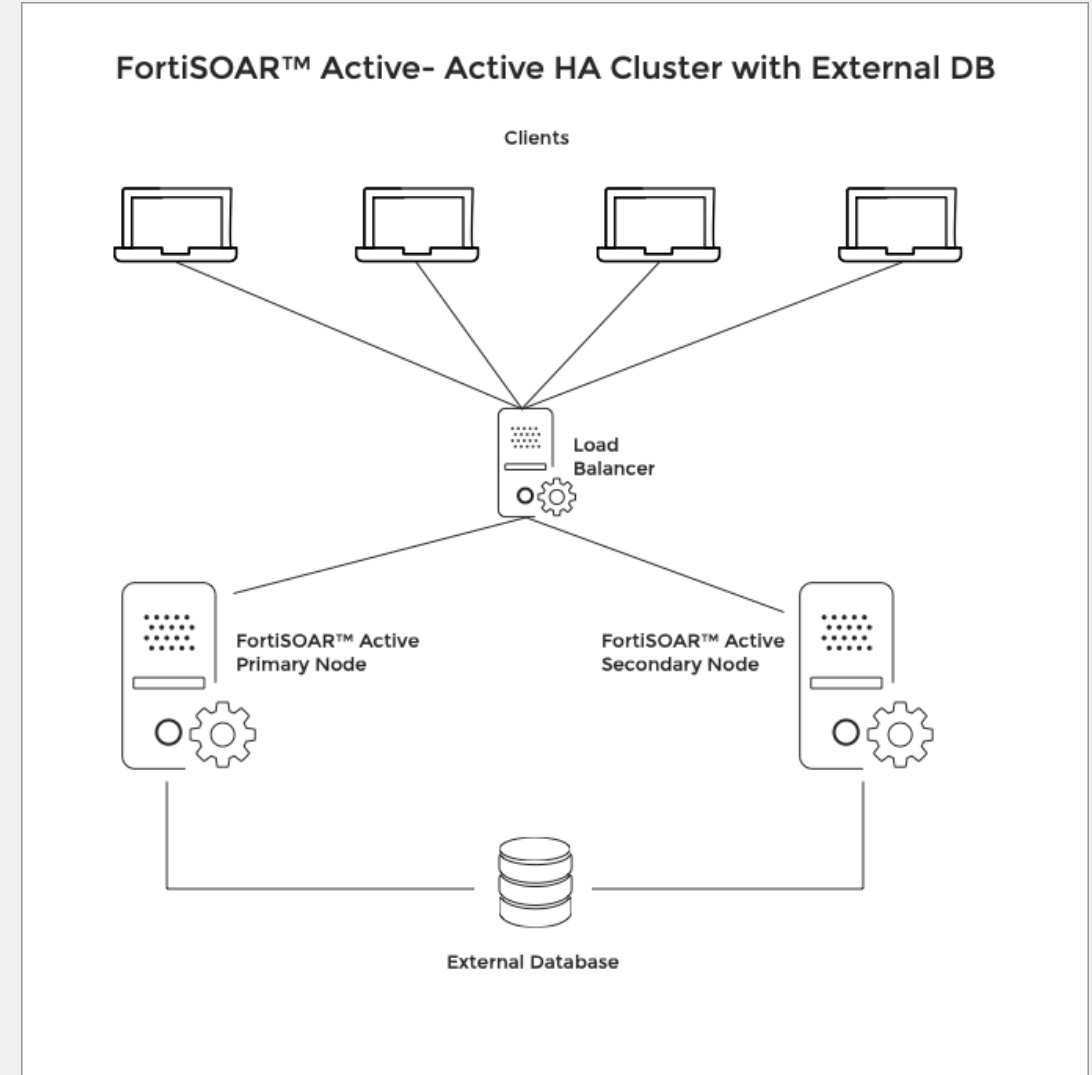
Configuring HA with an Internal Database

- Join nodes to the cluster in sequential order
- Connect to your VM as a root user and run the `csadm` command to configure HA
 - `csadm ha` command displays the options available to configure HA
- To configure a secondary node:
 - `csadm ha join-cluster --status <active, passive> --role secondary -- primary-node <Node_Name>`
- In case of a cloud environment with key-based authentication, you must use the following command:
 - `csadm ha join-cluster --status <active, passive> --role <primary, secondary> -- primary-node <Node_Name> --primarynode-ssh-key <Path_To_Pem_File>`
- The `list-nodes` command displays nodes in the cluster
 - When a node is in the process of joining the cluster, it will *not appear* in the `list-nodes` output



Configuring HA with an External Database

- Externalize the FortiSOAR database on the primary node first
- Add the host names of the secondary nodes to the `postgresql.conf` and `pg_hba.conf` files of the external database
- Ensure that you have met all HA prerequisites
- Create the HA cluster in a sequential order, one node at a time



HA Takeover

- Use the `csadm ha takeover` command to perform a take over when your active primary node is down
- Run the command on the secondary node that you want to configure as your active primary node
- Licenses are swapped between the old primary node and the new primary node
- During takeover if you answer 'no' at the Do you want to invoke 'join-cluster' on other cluster nodes? prompt, then you will have to reconfigure all nodes to point to the new active primary node

```
[root@secondary ~]# csadm ha takeover
Starting takeover operation
Promoting node as primary
Resetting the password for 'cyberpgsql' user
Enabling all services
Stopping all services
Starting all services
Clearing API cache
Awaiting 30 seconds for services to start up
Reindexing elasticsearch
Removing default database configuration file
Stopping all services
Starting all services
Successfully performed takeover
```

```
-----
Note:
Invoking 'join-cluster' on other cluster nodes will point those
cluster nodes to the new active primary node.
-----

Do you want to invoke 'join-cluster' on other cluster nodes? [yes/no]:yes
Checking whether postgresql server is running
Postgresql server is running
Replication user 'replicator' already exists
Checking whether postgresql server is running
Postgresql server is running
Stopping postgresql server
Successfully stopped postgresql server
Modifying settings in /var/lib/pgsql/12/data/postgresql.conf
Checking whether postgresql server is running
Postgresql server is not running
Starting postgresql server
Successfully started postgresql server
Opening port in firewall
```

HA Firedrill

- Tests your disaster recovery configuration when using an internal database
- Perform a firedrill on a secondary (active or passive) node only
 - `csadm ha firedrill`
- Suspends the replication to the node database
- Not applicable when the database is externalized
- After running the firedrill, run a restore operation to have the node rejoin the HA cluster
 - `csadm ha restore`
- All changes made while the node was in a firedrill will be discarded as test data

```
[root@primary ~]# csadm ha list-nodes
nodeId          nodeName          status  role    comment
-----
927ee24c7eabd15e684c996bd04ea968 secondary.internal.lab active  primary Performed Takeover at 2021-09-14 10:09:15AM
* 520e5f3badcd9647b829c8905fec465b primary.internal.lab  passive secondary Joined cluster with secondary.internal.lab
[root@primary ~]#
[root@primary ~]# csadm ha firedrill
Taking backup of postgresql configuration files
Taking backup of database configuration file
Starting firedrill operation
Promoting node as primary
Resetting the password for 'cyberpgsql' user
```

```
Taking backup of required files for re-joining cluster at [/opt/cyops-auth/.re-join-cluster.sql]
-----
Note:
Once you are done with testing, make sure to perform 'restore' operation
-----
Successfully performed firedrill
[root@primary ~]#
[root@primary ~]#
[root@primary ~]# csadm ha list-nodes
nodeId          nodeName          status  role    comment
-----
* 520e5f3badcd9647b829c8905fec465b primary.internal.lab active  primary Firedrill in progress. The operation was run at 2021-09-14 10:22:03AM
```

```
[root@primary ~]# csadm ha restore
Starting restore operation
Configuring as passive server
Validating rabbitmq credentials
Successfully validated rabbitmq credentials
Validating elasticsearch credentials
Successfully validated elasticsearch credentials
Validating postgresql credentials
```

```
Successfully configured as passive server
Removing firedrill backup directory
Successfully restored

root@primary ~]#
[root@primary ~]# csadm ha list-nodes
nodeId          nodeName          status  role    comment
-----
927ee24c7eabd15e684c996bd04ea968 secondary.internal.lab active  primary Performed Takeover at 2021-09-14 10:09:15AM
* 520e5f3badcd9647b829c8905fec465b primary.internal.lab  passive secondary Firedrill is complete. Joined cluster with secondary.internal.lab
[root@primary ~]#
```

Using the `csadm ha` Command

- The `csadm ha` command displays the options available to configure HA

Sub-command	Description
<code>list-nodes</code> <code>list-nodes --active</code> <code>list-nodes --active --secondary</code>	Lists all the nodes that are available in the cluster Retrieves nodes that are active Retrieves secondary active nodes
<code>export-conf</code>	Exports the configuration of the active primary node to a file named <code>ha.conf</code>
<code>allowlist</code>	Adds the hostnames of the secondary nodes to the allowlist on the active primary node
<code>join-cluster</code>	Adds a node to the cluster with the role and status you specify
<code>get-replication-status</code>	Displays the replication statistics. Only applicable to clusters with internal databases.
<code>show-health</code> <code>show-health --all nodes</code>	Displays the health information for the current node Displays the health information for all the nodes in an HA cluster
<code>firedrill</code>	Tests your disaster recovery configuration. Use it on a secondary (active or passive) node only
<code>restore</code>	Restores the node to its original state after you have performed a fire drill
<code>takeover</code>	Performs a takeover. Enter <code>csadm ha takeover</code> on the secondary node to configure it as your active primary node
<code>leave-cluster</code>	Removes a node from the cluster and the node goes back to the state it was in before joining the cluster

Knowledge Check

1. Which command checks the disaster recovery configuration on the secondary FortiSOAR instance?

A. `csadm ha show-health`

✓ B. `csadm ha firedrill`

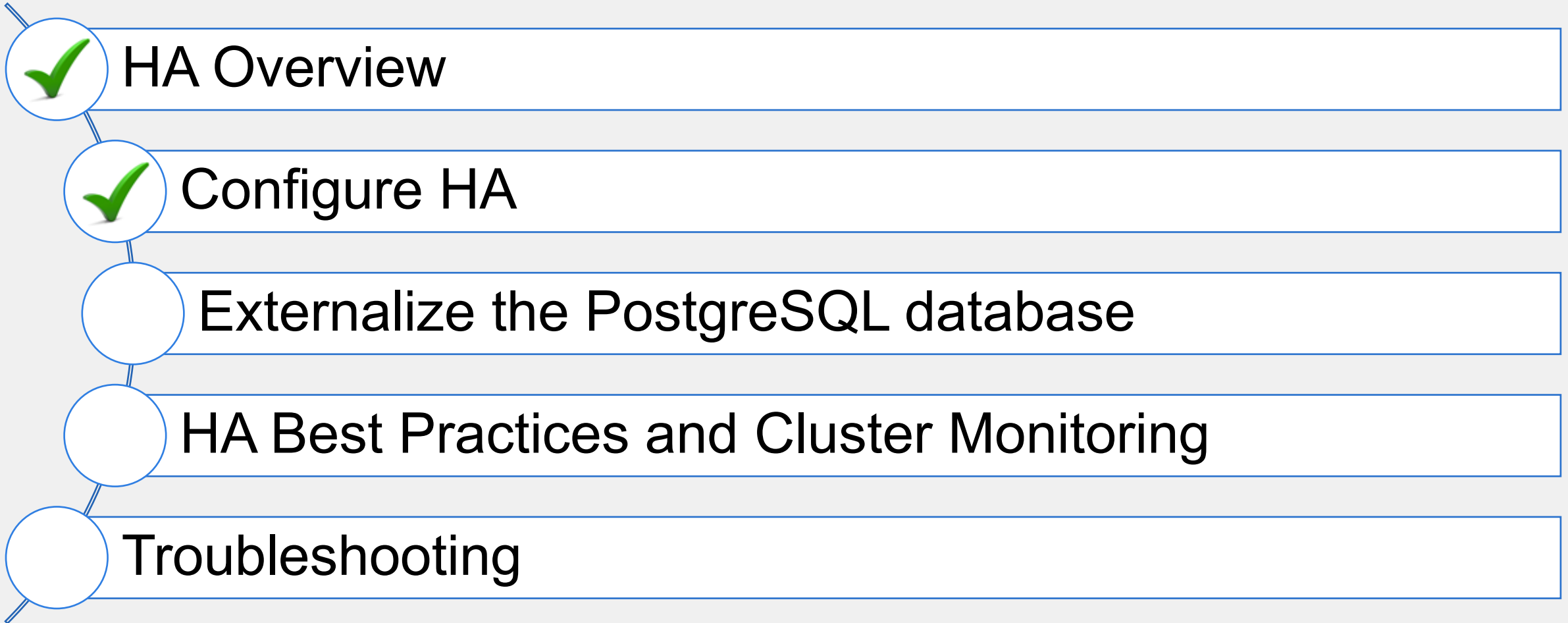
2. Which is a prerequisite for an HA cluster?

✓ A. All FortiSOAR nodes must be on the same version.

B. GUI must be accessible for cluster configuration.

C. All nodes require the same level of entitlement.

Lesson Progress



Externalize the FortiSOAR PostgreSQL database

Objectives

- Review the requirements for externalization of the PostgreSQL database
- Review the procedure to externalize the PostgreSQL database

Prerequisites for Externalization

PostgreSQL Instance

- Allow inbound connection from FortiSOAR
- Ensure PostgreSQL is version 14 (for 7.3.0)
- Release all stale connections:
`# systemctl restart postgresql-<postgresql version>`
- Verify that there is enough disk space (free disk space should at least be 3x the data size)
- Can run the following command to find out the current database size

```
# psadm db --getsize
```

FortiSOAR

- Verify connectivity with PostgreSQL instance
- Port 5432 needs to be open
- Stop all schedules and playbooks prior to externalizing

Externalizing FortiSOAR Databases

Step 1 Create the External Config File

- To create `db_external_config.yml`, copy the `db_config.yml` file in `/opt/cyops/configs/database/`

```
# cp /opt/cyops/configs/database/db_config.yml  
/opt/cyops/configs/database/db_external_config.yml
```

Step 2 Edit the External Config File

- `pg_external` parameter should be `true`
- Enter IP/DNS and port parameters of the PostgreSQL server
- Enter password set on server in the `pg_password` parameter
- To generate a hash for the password:
`csadm db -encrypt`
- SSL optional
- Archival settings can also be enabled and pointed to the same database server, but this is optional

```
postgres:  
  pg_external: true  
  pg_host: 10.0.1.170  
  pg_password:   
  pg_port: 5432  
  pg_ssl_cert_path: ''  
  pg_use_ssl: false  
  pg_user: cyberpgsql
```

Externalizing FortiSOAR Databases (Contd)

Step 3 Configure the Database Server

- Ensure that the server's firewall allows connections:

```
# firewall-cmd --add-service=postgresql -permanent  
# firewall-cmd --reload
```
- Edit the `pg_hba.conf` file to trust the FortiSOAR connections by adding these lines:

```
local all all trust  
host all all ip/subnetmask trust
```
- Configure `postgresql.conf` file to trust the FortiSOAR server for incoming connections, edit `/var/lib/pgsql/14/data/postgresql.conf` file:

```
listen_addresses = '*'  
port = 5432
```
- Restart the PostgreSQL service

```
# systemctl restart postgresql-14
```
- Create a `cyberpgsql` user:
 - ```
psql -U postgres -c "CREATE USER cyberpgsql WITH SUPERUSER
PASSWORD '<password>' "
```

# Externalizing FortiSOAR Databases (Contd)

## Step 4 Check Connectivity

- To check the connectivity between FortiSOAR and the database server  
`# csadm db --check-connection`

## Step 5 Initiate Externalization

- To start the externalize process  
`# csadm db --externalize`

## Step 6 Restart Schedules

- After externalizing is complete, restart all schedules and playbooks

# Knowledge Check

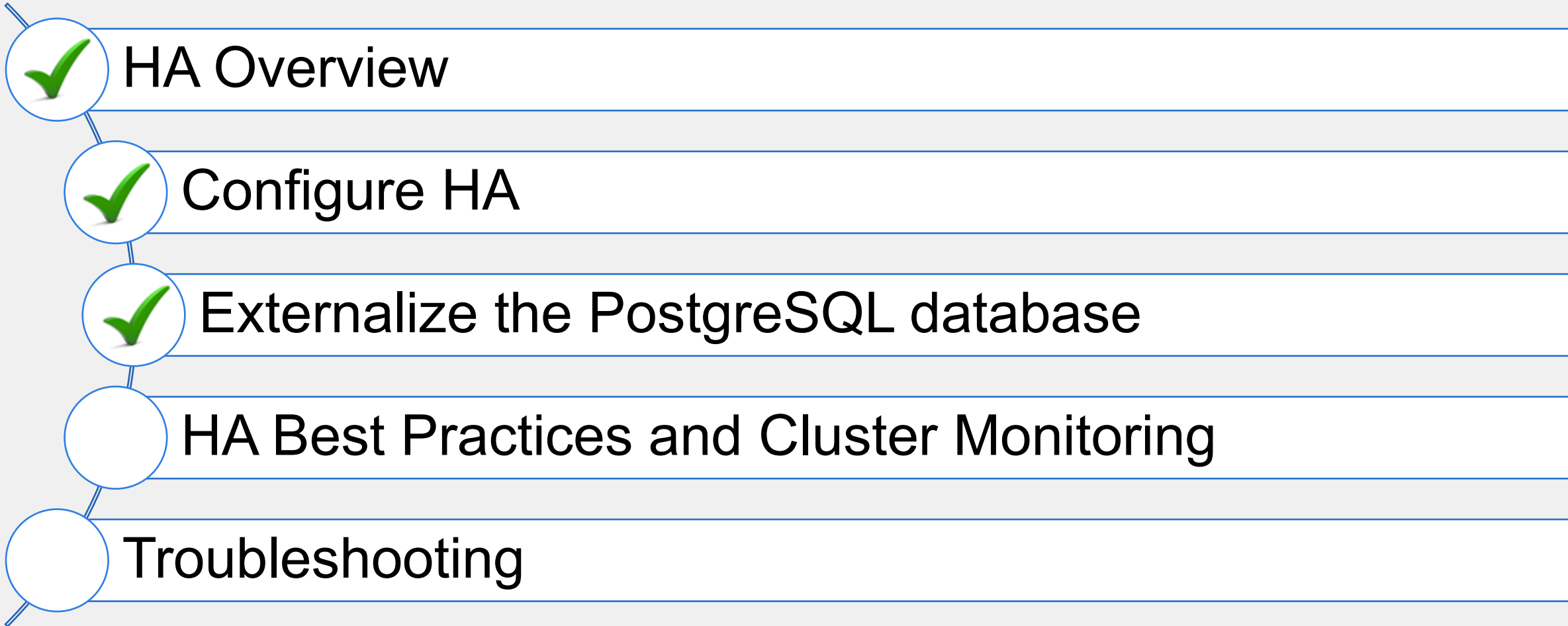
1. Which version of the PostgreSQL database is supported for externalization?

A. 11



B. 14

# Lesson Progress



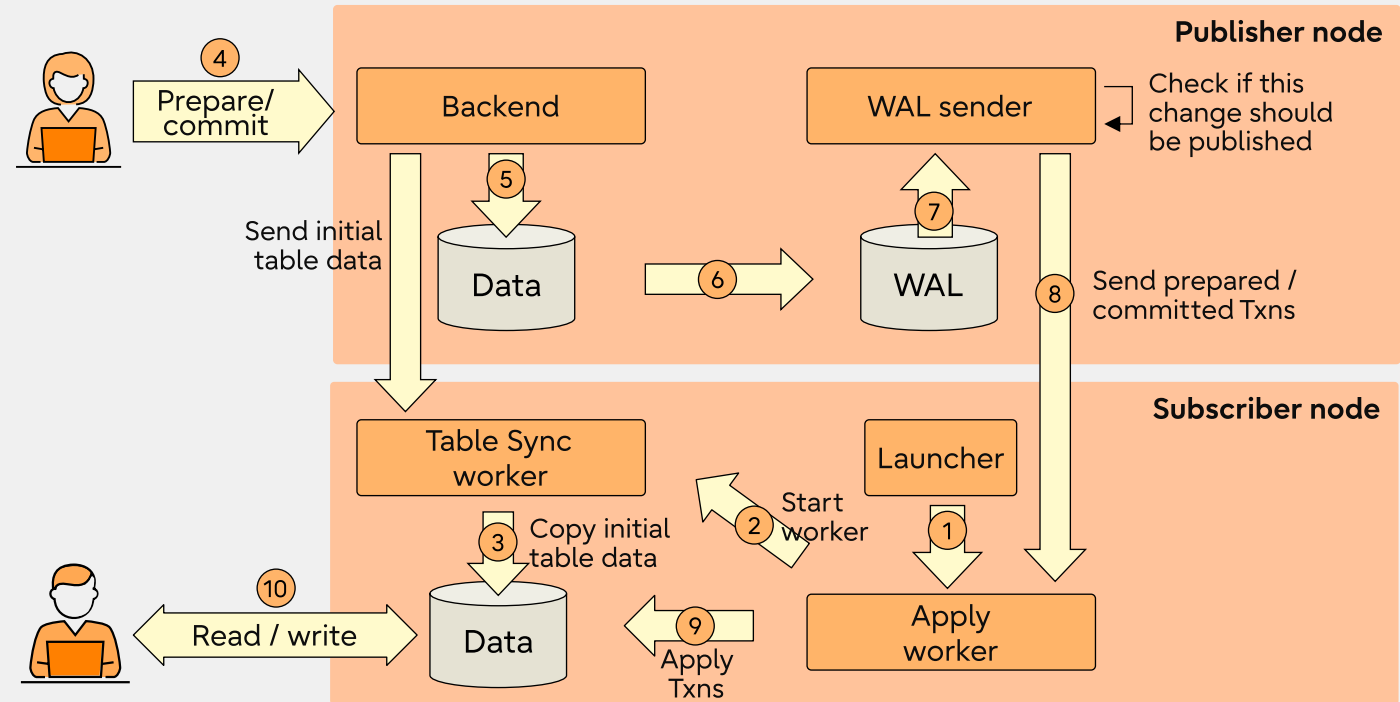
# HA Best Practices and Cluster Monitoring

## Objectives

- Identify HA best practices
- Fine-tune the HA configuration
- Set up HA cluster monitoring
- Understand the HA cluster notifications checks

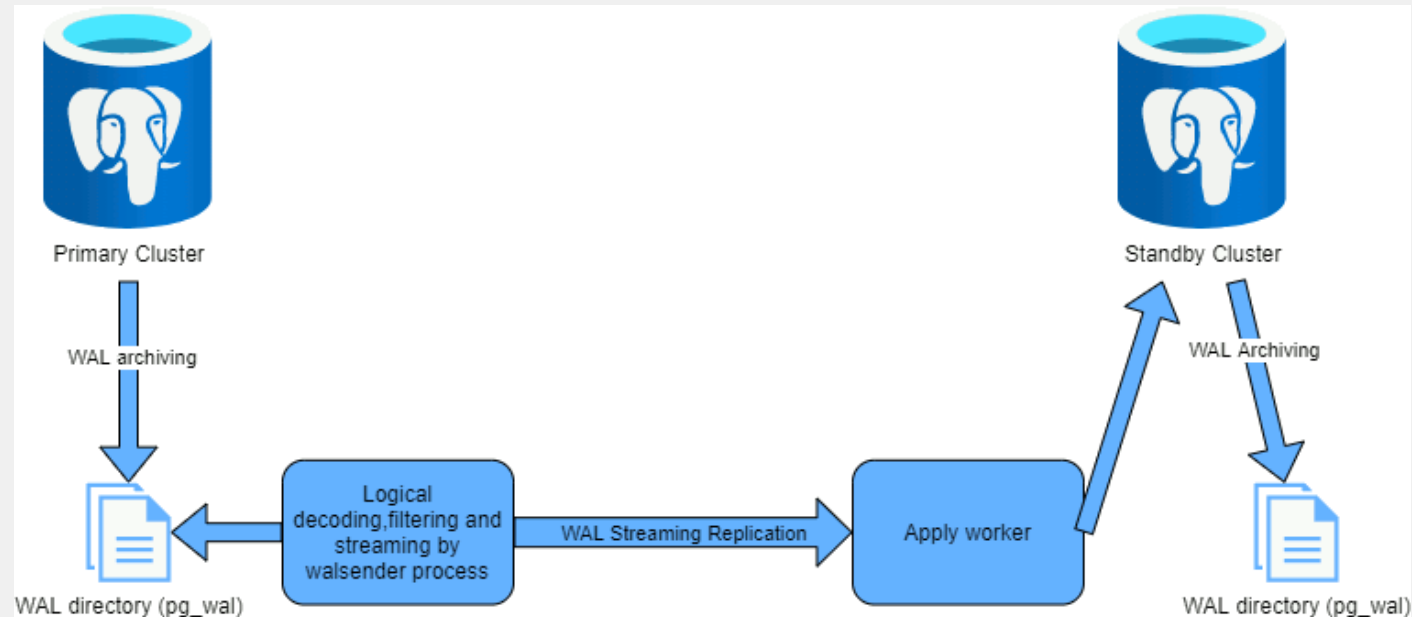
# HA Best Practices

- Deploy the cluster behind a load balancer so that the address remains unchanged on failover
- Ensure endpoints are reachable using DNS even after a failover
- The pull model ensures resiliency
  - FortiSOAR node connects to outbound SIEM periodically to pull alerts
  - In case of downtime, once the FortiSOAR node comes back up, it pulls the alerts from the last pulled time, ensuring there is no data loss even during the outage



# Fine-Tuning the HA Configuration

- Setting changes that you make in any configuration file on an instance apply only to that instance.
  - Must make those changes across all the cluster nodes manually
- There are two attributes you can configure in the `postgresql.conf` file to fine-tune a FortiSOAR HA:
  - **max\_wal\_senders**
    - Defines the maximum number of `walsender` processes
  - `max_replication_slots`
    - Defines the maximum number of replication slots
- Every secondary node needs one **walsender** process and one replication slot on the primary node
- If more than 10 secondary nodes, increase the value of the `max_wal_senders` and `max_replication_slots` attributes in the `postgresql.conf` file on the primary node and restart the PostgreSQL server





# Fine-Tuning the HA Configuration (Contd)

- Shared memory settings in the `postgresql.conf` file on the secondary nodes should be at least equal to the primary node
- The parameters used for tracking transactional IDs, locks, and prepared transactions are:
  - `max_connections`
  - `max_prepared_transactions`
  - `max_locks_per_transaction`
  - `max_wal_senders`
  - `max_worker_processes`
- If the secondary nodes have lower values than the primary node, HA operations could run into potential issues if there are not enough resources

# Cluster Health Monitoring

- **Monitoring Interval (Minutes):** specifies the interval at which you want to monitor the system and perform the health check of the HA cluster
- **Missed Heartbeat Count:** specifies the count of missed heartbeats.
- **Replication Lag:** specifies the threshold for the replication lag between nodes
- If **Monitoring Interval** is set to **5** minutes and the **Missed Heartbeat Count** to **3**, this means that when the heartbeat is missed for the last 15 minutes or more, the heartbeat missed notification is sent to the email address that you specify in the **Email** field

## Settings > System Configurations > Application Configuration

The screenshot shows the 'Application Configuration' page for the 'SMTP' service. The 'Email' field is set to 'csadmin@internal.lab'. The 'Monitoring Interval (Minutes)' is set to '5'. The 'Missed Heartbeat Count' is set to '3'. The 'Replication Lag (GB)' is set to '3'. The 'System Health Thresholds' section includes settings for Memory Utilization (80%), CPU Utilization (80%), Disk Utilization (80%), Swap Memory Utilization (50%), Workflow Queue Threshold (100), and WAL Files Size (20 GB). The 'Cluster Health' section includes a note about missed heartbeats and replication lag.

| Service | Email                |
|---------|----------------------|
| SMTP    | csadmin@internal.lab |

Please ensure your SMTP service is configured.

**Monitoring Interval (Minutes)**  
5  
The monitoring job will run at this schedule

**System Health Thresholds**  
A notification will be generated when the resource consumption on the server is high.  
Define the respective thresholds below:

| Memory Utilization (%) | CPU Utilization (%) | Disk Utilization (%) |
|------------------------|---------------------|----------------------|
| 80                     | 80                  | 80                   |

| Swap Memory Utilization (%) | Workflow Queue Threshold | WAL Files Size (GB) |
|-----------------------------|--------------------------|---------------------|
| 50                          | 100                      | 20                  |

**Cluster Health**  
An email notification will also be generated if there are missed heartbeats from any of the cluster nodes, or if the replication lag is high.  
Define the respective thresholds below:

| Missed Heartbeat Count | Replication Lag (GB) |
|------------------------|----------------------|
| 3                      | 3                    |

Select the service to be used for notification: **SMTP** or **Exchange**

Available when node is part of an HA cluster

# Understanding HA Cluster Notification Checks

- HA cluster failure email notifications provide useful information about potential causes, and also remediation steps:

Following secondary FortiSOAR node(s) **seem to have failed** -

Node: **hasecondary.myorgdomain**

Current lag with primary node is 97 kB

Failure reason:

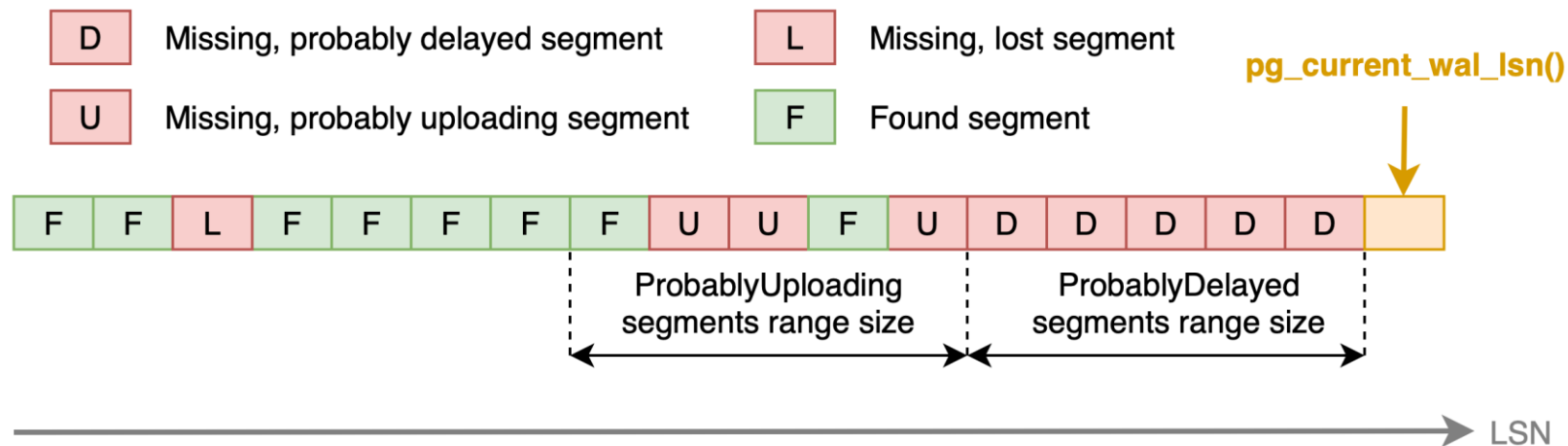
1. The postgres database replication to the secondary node is not working due to data log rotation at the primary node.
2. The secondary node has been shutdown/halted.
3. PostgreSQL not running on node(s).
4. nodeName from 'csadm ha list-nodes' differs from actual FQDN used during join-cluster.

If node is up and running,

1. Check the status of PostgreSQL service using 'systemctl status postgresql-<postgresql-version-here> -l' on node to get more details.
2. If you see 'FATAL: could not receive data from WAL stream: requested WAL segment has already been removed' in the PostgreSQL service status, you need to re-join the cluster using '**csadm ha join-cluster --fetch-fresh-backup**'

# Understanding HA Cluster Notification Checks (Contd)

- If the replication lag reaches or crosses the threshold specified:  
Replication lag threshold is reached on following node(s):  
Node: hasecondary.myorgdomain  
Current lag with primary node : 3431 MB  
Configured Threshold : 3 GB
- If any services are not running, then the health notification check sends a “service failure” notification and exits



# Understanding HA Cluster Notifications (Contd)

- If a firedrill is in progress on a secondary node, then the health notification check sends the following notification and exits:

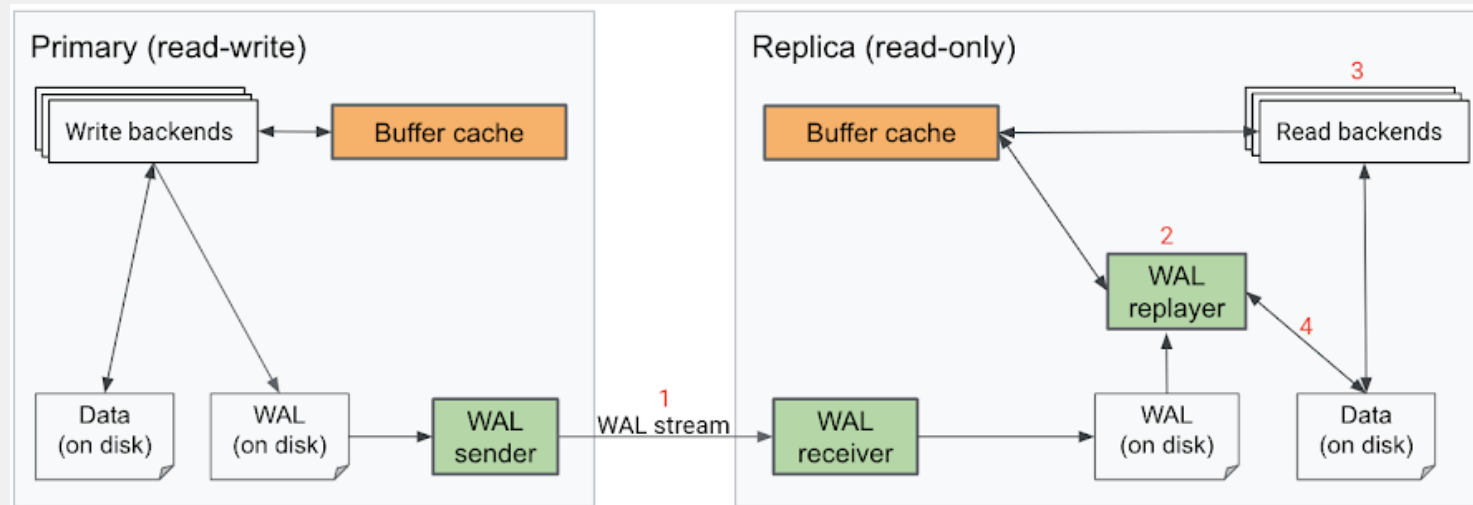
Firedrill is in progress on following node(s):

Node: hasecondary.myorgdomain

Current lag with primary node : 52 kB

You can ignore this lag

- You can ignore the lag that the system displays in this case because this lag indicates the amount of data the fire drill node needs to sync when you run the `csadm ha restore` command

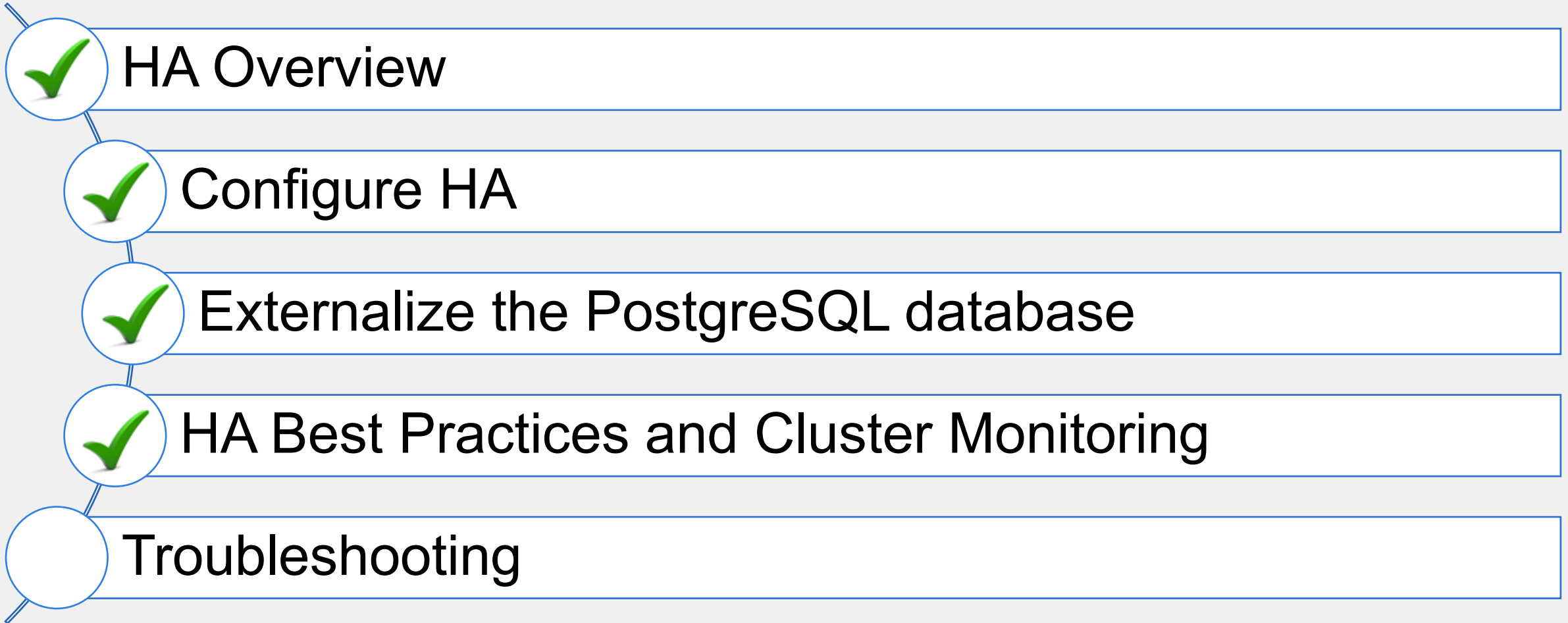


# Knowledge Check

1. How many secondary/passive nodes can you configure in an HA cluster before you must change the `max_wal_sender` attribute?

- ✓ A. 10
- B. 15

# Lesson Progress



# Troubleshooting

## Objectives

- Troubleshoot issues based on notifications
- Troubleshoot PostgreSQL HA replication issues
- Troubleshoot failure to create HA issues



# Troubleshooting Issues Based on Notifications

- To troubleshoot HA issues, review logs in `/var/log/cyops/cyops-auth/ha.log`
- Some example issues:

## 1. Heartbeat failure

- Check if the cyops-ha service is running
  - `systemctl status cyops-ha`
- If it is not running, then start the cyops-ha service

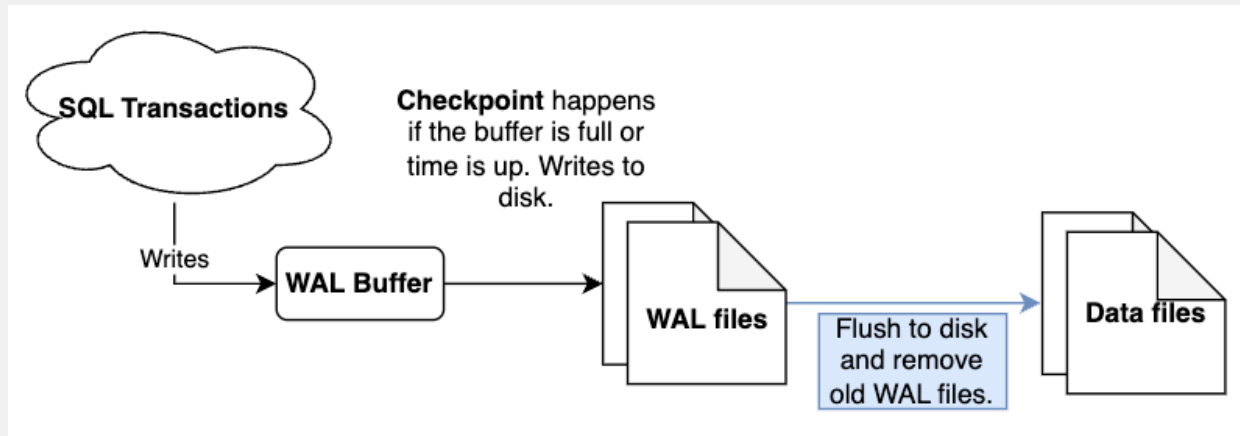
## 2. Node name differs from actual FQDN—nodeName from 'csadm ha list-nodes' differs from actual FQDN used during join-cluster

- Log in to the node and run
  - `csadm hostname --set <enter-correct-FQDN-here>`

# Troubleshooting Issues Based on Notifications (Contd)

## 3. Secondary node is out-of-sync with the primary node

- PostgreSQL service status shows requested WAL segment has already been removed, or the `csadm ha get-replication-stat` command shows a higher time lapsed from the last sync when compared to the general time lapsed
- In either case, the secondary node is completely out-of-sync with the primary node. Perform the following steps on the secondary node:
  - Create the `.joincluster_in_progress` file: `touch /home/csadmin/.joincluster_in_progress`
  - Rejoin the cluster: `csadm ha join-cluster --status active/passive --role secondary --primary-node <Primary-node-FQDN> --fetch-fresh-backup`
- When there are heavy write operations on the primary node and the secondary node has not yet copied the data before the data has rolled over, the primary and secondary nodes are out of sync and need to be fully synchronized, which can cause the above failures
  - Increase the `wal_keep_size` value in the `/var/lib/pgsql/14/data/postgresql.conf` file
  - Restart the service: `systemctl restart postgresql-14`



# Troubleshooting Issues Based on Notifications (Contd)

## 4. PostgreSQL service is down on the primary node or external database server

- As a result, `cyops-ha` service will be down across the entire cluster
- You will not be able to access the FortiSOAR GUI
- Check the PostgreSQL server status by running
  - `systemctl status postgresql-14 -l`
- Identify the root cause for the service being down, and then restore services
- Can try to restart the service by running
  - `systemctl start postgresql-14`

# Troubleshooting PostgreSQL Replication Issues

- The PostgreSQL service on the secondary/passive node has stopped, but is still running on the primary
- Run the `csadm ha get-replication-stat` command on the primary node:

```
csadm ha get-replication-stat
Warning: Following could be the issues with the nodes:
1. The postgres database replication to the secondary node is not working due to data log rotation at the primarynode.
2. The secondary node has been shutdown/halted.
3. PostgreSQL not running on node(s).
4. nodeName from 'csadm ha list-nodes' differs from actual FQDN used during join-cluster.
5. If a firedrill is in progress on the node, no action is required. The 'lag' that is displayed indicates the amount of data the firedrill node needs to sync when 'csadm ha restore' will be performed.
```

If node is up and running,

1. Check the status of PostgreSQL service using `'systemctl status postgresql-14 -l'` on node to get more details.
2. If you see `'FATAL: could not receive data from WAL stream: requested WAL segment has already been removed'` in the PostgreSQL service status, you need to re-join the cluster using `'csadm ha join-cluster --fetch-fresh-backup'` for this node.

| nodeId                      | nodeName                | status | role      | comment                                   | total_lag |
|-----------------------------|-------------------------|--------|-----------|-------------------------------------------|-----------|
| 469c633061c30dd4d8e3a607cf2 | hasecondary.myorgdomain | active | secondary | Joined cluster with haprimary.myorgdomain | 4702 MB   |

Secondary node is lagging with 4702 MB

# Troubleshooting PostgreSQL Replication Issues (Contd)

- PostgreSQL service on the secondary/passive node is resumed, and the replication stat is observed on the primary and secondary node:
- On the primary node:

```
csadm ha get-replication-stat
```

Note:

'**sending\_lag**' indicates load on the primary node

'**receiving\_lag**' indicates network delay or load on the passive/secondary node

'**replaying\_lag**' indicates load on the passive/secondary node

| node_hostname        | sending_lag | receiving_lag | replaying_lag | total_lag |
|----------------------|-------------|---------------|---------------|-----------|
| -----                | -----       | -----         | -----         | -----     |
| hasecondary.mydomain | 4458 MB     | 11 MB         | 213 MB        | 4683 MB   |

- On the secondary node:

| primary_hostname   | total_lag | time_elapsed_from_last_sync |
|--------------------|-----------|-----------------------------|
| -----              | -----     | -----                       |
| haprimary.mydomain | 287 MB    | 00:07:59.113185             |

Secondary node is lagging with 4683 MB

# Troubleshoot—Failure to Create an HA Cluster

- If the process to configure HA using the automated `join-cluster` fails
  - As root, run the `csadm ha` command to display a list of available options
- To configure a node as a secondary node:
  - Connect over SSH to the active primary node and run the `csadm ha export-conf` command to export the configuration details of the primary node to a file named `ha.conf`
  - Copy the `ha.conf` file from the active primary node to the node that you want to configure as a secondary node
  - On the primary node, add the host names of the secondary nodes to the allowlist:
    - `csadm ha allowlist --nodes hostnameA, hostnameB`
  - In the case of an externalized database, you must add all the nodes in the cluster to the allowlist in the `pg_hba.conf` file
  - Ensure that all HA nodes are resolvable through DNS
  - SSH to the server that you want to configure as a secondary node, and run the following command:
    - `csadm ha join-cluster --status <active, passive> --role <primary, secondary> --conf <location of the ha.conf file>`
  - If you run the `csadm ha join-cluster` command without adding the host names of the secondary nodes to the allowlist, then you will get a `Failed to verify` error
  - When you join a node to an HA cluster, the `list-nodes` command does not display that a node is in the process of joining the cluster. The newly added node will be displayed only after it has joined the cluster

# Troubleshoot—Primary Node Is Faulted

- If the primary node is in a **Faulted** state, it could be because there was a system crash and the administrator triggered a failover
- After the primary node resumes running, it still thinks it is the primary node in its own cluster
- Other nodes display the primary node in a **Faulted** state
- To resolve this issue:
  1. After the former primary node resumes running, run the `leave-cluster` command on it, which will remove this node from the HA cluster
  2. Run the `join-cluster` command to join this node back to the HA cluster with the new primary node

Settings > License Manager

| Nodes                  |         |           |                                                                                                                                                                                          |
|------------------------|---------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Node Name              | Status  | Role      | License Details                                                                                                                                                                          |
| primary.internal.lab   | Faulted | Secondary | Serial Number: FSRVMPTM20000484<br>Total Users: 2<br>Device UUID: 520e5f3badcd9647b829c8905fec465b  |
| secondary.internal.lab | Active  | Primary   | Serial Number: FSRVMPTM21000105<br>Total Users: 2<br>Device UUID: 927ee24c7eabd15e684c996bd04ea968  |

# Knowledge Check

1. In the output of the `csadm ha get-replication-stat` command, what does `replaying_lag` on the primary FortiSOAR indicate?
  - A. Load on the primary node
  - ✓ B. Load on the secondary node
2. To fix a former primary node in a faulted state, which action should you perform?
  - ✓ A. Leave the cluster and rejoin it.
  - B. Initiate an HA takeover to become the primary node again.



# Lesson Progress

- ✓ HA Overview
- ✓ Configure HA
- ✓ Externalize the PostgreSQL database
- ✓ HA Best Practices and Cluster Monitoring
- ✓ Troubleshooting

# Review

- ✓ Identify different HA implementation prerequisites and configuration options
- ✓ Manage cluster licensing
- ✓ Configure HA with an internal or external PostgreSQL database
- ✓ Identify HA best practices
- ✓ Set up cluster monitoring
- ✓ Understand HA cluster notifications checks
- ✓ Troubleshoot HA issues